Applied Statistics 307312
Course Syllabus

Master's level | Student Form | Draft

Course information	Details
Course no. / title	307312 / Applied Statistics
Academic level	Master's (graduate)
Academic year / semester	[To be confirmed]
Credit hours / delivery mode	[Confirm from approved course specification]
Prerequisite / co-requisite	[To be confirmed]
Instructor	Dr. Mohammed Fasha [confirm assignment]
Email / office	mohammed.fasha@uop.edu.jo / [confirm office]
LMS	Moodle: https://lms.uop.edu.jo
Office hours	[Announce on Moodle]

Course Description

This graduate-level course develops applied statistical reasoning for business and research decisions. Students prepare and visualize data; describe distributions and associations; use probability, sampling, estimation, hypothesis testing, and group comparisons; and fit introductory regression and forecasting models. R is the analysis language, and Quarto combines code, tables, charts, and interpretation in reproducible reports. The course follows seven topic modules and uses selected textbook examples with documented data sources.

Course Objectives

- Develop statistically sound approaches to collecting, preparing, describing and visualizing data.
- Apply probability and sampling theory to quantify variability and uncertainty.
- Select, conduct and interpret estimation, hypothesis testing and comparison procedures.
- Build, assess and communicate regression and introductory forecasting models.
- Use R and Quarto to produce reproducible analyses and communicate defensible evidence for business decisions.

Course Intended Learning Outcomes (ILOs)

ILO	Outcome	Teaching and assessment
K1	Explain variable types, sampling, descriptive summaries, probability models and sampling distributions.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report
K2	Evaluate the logic, assumptions and interpretation of estimation, hypothesis testing and statistical models.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report
I1	Select appropriate statistical procedures for a stated business or research question and justify choices.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report
I2	Critically evaluate statistical evidence, uncertainty, model assumptions and limitations.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report
P1	Prepare, visualize and summarize real-world datasets using R.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report

ILO	Outcome	Teaching and assessment
P2	Implement inferential procedures, regression analyses and basic forecasts using R.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report
C1	Communicate statistically defensible findings in a reproducible Quarto report for a nontechnical audience.	Concept discussion; guided R practice; quizzes or examinations; reproducible Quarto report

Formal alignment of the course outcomes with the approved Master's program outcomes remains subject to departmental confirmation.

Course Schedule (15 Teaching Weeks)

Week	Topic(s)	Topic details	Teaching / student tasks
1	Module 1: Introduction to R	R/RStudio setup; scripts and projects; objects, vectors, data frames; importing CSV; packages; basic functions and visualizations; reproducible workflows. Reading: Introductory R and Quarto readings.	Set up R and Quarto; inspect a CSV; run a script; render a short report.
2	Module 2: Data and descriptive statistics	Research questions, data sources, populations and samples, measurement scales, cleaning, frequency tables and charts. Reading: Main book Chapters 1–2.	Classify Restaurant variables; summarize SoftDrink purchases and Audit durations using tables and charts.
3	Modules 2–3: Numerical measures and basic probability	Location and dispersion, quantiles, association; events, conditional probability, independence and Bayes theorem. Reading: Main book Chapters 3–4.	Calculate Audit summaries and Electronics association; complete the descriptive review and introduce probability.
4	Module 3: Discrete and continuous distributions	Binomial and Poisson models; normal, uniform and exponential distributions; probabilities and quantiles. Reading: Main book Chapters 5–6.	Evaluate, simulate and plot probability distributions in R.
5	Module 4: Sampling and estimation	Sampling variability, central limit theorem, standard errors; confidence intervals for means and proportions. Reading: Main book Chapters 7–8.	Simulate sampling distributions and estimate confidence intervals in R.
6	Module 5: One-sample tests	Hypotheses, p-values, significance, errors, assumptions, one-sample t and proportion tests. Reading: Main book Chapter 9.	Conduct and interpret one-sample tests in R.
7	Module 5: Two-sample tests	Independent and paired comparisons; two-sample t tests and proportion tests. Reading: Main book Chapter 10.	Compare two groups using R and report effect size and uncertainty.
8	Module 5: ANOVA, chi-square, and rank-based methods	One-way ANOVA and post-hoc analysis; chi-square goodness-of-fit and independence; rank-based tests. Reading: Main book Chapters 12–13; selected nonparametric notes.	Run ANOVA, chi-square and selected nonparametric tests in R.
9	Module 6: Simple linear regression	Least squares; interpretation, residual diagnostics; confidence and prediction intervals. Reading: Main book Chapter 14.	Fit and diagnose a simple regression model in R.
10	Module 6: Multiple regression	Partial effects, multiple predictors, dummy variables, model fit and coefficient interpretation. Reading: Main book Chapter 15.	Estimate and interpret a multiple regression model in R.
11	Module 6: Model building and validation	Interactions, multicollinearity, residual diagnostics, model selection and validation. Reading: Main book Chapter 15 and validation notes.	Build, compare and validate regression models in R.

Week	Topic(s)	Topic details	Teaching / student tasks
12	Module 7: Time-series forecasting	Trend, seasonality, moving averages, baseline forecasts and forecast error. Reading: Selected forecasting readings.	Generate and evaluate a short time-series forecast in R.
13	Module 7: Business analytics	Decision framing, descriptive and predictive analytics, communicating findings and responsible use of evidence. Reading: Main book Chapter 1 and assigned case.	Prepare an evidence-based business analytics case using R.
14	Module 7: Reproducibility and responsible analysis	Reproducibility, ethical data analysis, analytical planning and emerging workflows. Reading: Main book Section 1.9 and reproducibility notes.	Submit a reproducible analysis and reflect on limitations and responsible AI use.
15	Integration and review across modules	Integrate data preparation, exploratory analysis, statistical inference and modeling; final assessment preparation. Reading: Selected review exercises.	Present and review an integrated R analysis; practice interpreting statistical outputs.

The schedule follows seven course modules. A module may span several weeks, and Week 3 bridges descriptive statistics and probability. Module numbers identify course topics, not textbook chapters. Timing and examination dates remain subject to the approved academic calendar and contact hours.

Assessment Methods and Grading

Assessment	Proposed weight	Evidence
Assignments, R exercises and quizzes	30%	Individual applied problems and reproducible code
Midterm examination	30%	Conceptual and analytical problem solving
Final examination	40%	Integrated statistical analysis and interpretation
Total	100%	Subject to departmental approval

The proposed distribution is 30% coursework, including assignments, R exercises and quizzes; 30% midterm examination; and 40% final examination, subject to departmental approval. A reproducible Quarto report may contribute to coursework when assigned; it does not add a separate assessment weight.

Learning References and Resources

Main statistical reference: assigned readings, printed page references, and the selected textbook cases use the following edition.

Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., Cochran, J. J., Fry, M. J., and Ohlmann, J. W. (2020). Essentials of Modern Business Statistics with Microsoft Excel (8th ed.). Cengage Learning. ISBN 978-0-357-13162-6.

Available at the university library.

Supplementary statistical reference: use when assigned; it does not replace the main 8th-edition book.

Berenson, M. L., Levine, D. M., and Krehbiel, T. C. (2004). Basic Business Statistics: Concepts and Applications (9th ed.). Pearson Prentice Hall. Supplementary reading only; its chapter and dataset numbering differs from the main reference.

Available at the university library.

R reference: Wickham, H., Çetinkaya-Rundel, M., and Grolemund, G. R for Data Science (2nd ed.). Free online: https://r4ds.hadley.nz/

Introductory R reference: Grolemund, G. (2014). Hands-On Programming with R. Free online: https://rstudio-education.github.io/hopr/

Quarto guidance: Get Started and R tutorials, https://quarto.org/docs/get-started/.

Software: R, RStudio or an equivalent R environment, and Quarto. Core practical work uses ggplot2, dplyr, tidyr, readr, and knitr; further packages are introduced as needed. Student analyses read the prepared CSV files. Original publisher spreadsheets are retained as source records, and checks against the chosen edition are documented. Textbook Excel instructions are adapted to R rather than requiring Excel as the analysis platform.

Course materials: follow the numbered modules and their practice in the course website and Moodle. Use Restaurant for variable meaning and group comparisons, SoftDrink for categorical frequencies, Audit for distributions and numerical summaries, and Electronics for numerical association. Supplementary illustrations are labeled separately. Assigned dataset and figure sources must remain attributed.

Course Modules and Reading Guide

No.	Topic	Selected reading
1	Introduction to R	R for Data Science (2e), selected data import, transformation, visualization, and Quarto sections; introductory R notes.
2	Data and descriptive statistics	Main book Chapter 1, selected sections; Chapters 2–3. Restaurant, SoftDrink, Audit, and Electronics examples.
3	Probability and distributions	Main book Chapters 4–6, selected sections.
4	Sampling and estimation	Main book Chapters 7–8, selected sections.
5	Hypothesis testing and group comparisons	Main book Chapters 9–10, Chapter 12 (chi-square), and Chapter 13 (ANOVA); supplementary readings for rank-based methods.
6	Regression	Main book Chapters 14–15, selected sections; supplementary readings for advanced validation.
7	Forecasting and applied analysis	Instructor-selected forecasting and applied-case readings; Chapter 1 for business analytics and responsible statistical practice.

Course Requirements and Policies

• Students need computer and Moodle access and an R environment capable of rendering Quarto documents. Follow the setup guide and preserve the course project folder structure.

• Attendance, examination and academic-integrity policies follow current University of Petra regulations.

• Submitted work must identify sources, report methods transparently and distinguish individual work from permitted collaboration.

• AI use and disclosure: Students must declare any use of generative AI or other AI tools in assignments, projects, reports, code and other submitted coursework, whether or not AI use is expressly permitted. The declaration must name the tools used, describe how they were used, and identify affected sections or outputs. If no AI tools were used, students must state “No AI tools were used.” Undeclared or prohibited use is subject to University of Petra academic-integrity regulations; students remain responsible for the accuracy, originality and integrity of all submissions.

Assessment and Submission Guidance

Submit the Quarto source (.qmd) and its rendered report unless an assignment specifies otherwise. Include the dataset/source, reproducible R code, labeled tables and plots, interpretation, limitations, and the required AI-use declaration. Keep input CSVs unchanged. Submission dates, examination coverage, and rubrics will be announced through Moodle.

Approval / Institutional Use

Role	Name / signature / date
Curriculum committee coordinator	[Pending]
Department head / Dean	[Pending]